

Worst, Average and Best Case Analysis of Algorithms

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In the [previous post](#), we discussed how Asymptotic analysis overcomes the problems of the naive way of analyzing algorithms. Now let us learn about What is Worst, Average, and Best cases of an algorithm:

1. Worst Case Analysis (Mostly used)

- In the worst-case analysis, we calculate the upper bound on the running time of an algorithm. We must know the case that causes a maximum number of operations to be executed.
- For [Linear Search](#), the worst case happens when the element to be searched (x) is not present in the array. When x is not present, the `search()` function compares it with all the elements of `arr[]` one by one.
- This is the most commonly used analysis of algorithms (We will be discussing below why). Most of the time we consider the case that causes maximum operations.

2. Best Case Analysis (Very Rarely used)

- In the best-case analysis, we calculate the lower bound on the running time of an algorithm. We must know the case that causes a minimum number of operations to be executed.
- For [linear search](#), the best case occurs when x is present at the first location. The number of operations in the best case is constant (not dependent on n). So the order of growth of time taken in terms of input size is constant.

3. Average Case Analysis (Rarely used)

- In average case analysis, we take all possible inputs and calculate the computing time for all of the inputs. Sum all the calculated values and divide the sum by the total number of inputs.

- We must know (or predict) the distribution of cases. For the linear search problem, let us assume that all cases are uniformly distributed (including the case of x not being present in the array). So we sum all the cases and divide the sum by $(n+1)$. We take $(n+1)$ to consider the case when the element is not present.

Why is Worst Case Analysis Mostly Used?

Average Case : The average case analysis is not easy to do in most practical cases and it is rarely done. In the average case analysis, we need to consider every input, its frequency and time taken by it which may not be possible in many scenarios

Best Case : The Best Case analysis is considered bogus. Guaranteeing a lower bound on an algorithm doesn't provide any information as in the worst case, an algorithm may take years to run.

Worst Case: This is easier than average case and gives an upper bound which is useful information to analyze software products.

Interesting information about asymptotic notations:

A) For some algorithms, all the cases (worst, best, average) are asymptotically the same. i.e., there are no worst and best cases.

- **Example:** Merge Sort does order of $n \log(n)$ operations in all cases.

B) Where as most of the other sorting algorithms have worst and best cases.

- **Example 1:** In the typical implementation of Quick Sort (where pivot is chosen as a corner element), the worst occurs when the input array is already sorted and the best occurs when the pivot elements always divide the array into two halves.
- **Example 2:** For insertion sort, the worst case occurs when the array is reverse sorted and the best case occurs when the array is sorted in the same order as output.

Examples with their complexity analysis:

1. Linear search algorithm:

C++

C

Java

Python

C#

JavaScript

```
#include <iostream>
#include <vector>
using namespace std;

// Linearly search target in arr.
// If target is present, return the index;
// otherwise, return -1
int search(vector<int>& arr, int x) {
    for (int i = 0; i < arr.size(); i++) {
        if (arr[i] == x)
            return i;
    }
    return -1;
}

int main() {
    vector<int> arr = {1, 10, 30, 15};
    int x = 30;
    cout << search(arr, x);
    return 0;
}
```

Output

2

Best Case: Constant Time irrespective of input size. This will take place if the element to be searched is on the first index of the given list. So, the number of comparisons, in this case, is 1.

Average Case: Linear Time, This will take place if the element to be searched is at the middle index (in an average search) of the given list.

Worst Case: The element to be searched is not present in the list

2. Special Array Sum : In this example, we will take an array of length (n) and deals with the following cases :

- If (n) is even then our output will be 0
- If (n) is odd then our output will be the sum of the elements of the array.

Below is the implementation of the given problem:

C++

C

Java

Python

C#

JavaScript

PHP

```
#include <iostream>
#include <vector>
using namespace std;

int getSum(const vector<int>& arr1) {
    int n = arr1.size();
    if (n % 2 == 0) // (n) is even
        return 0;

    int sum = 0;
    for (int i = 0; i < n; i++) {
        sum += arr1[i];
    }
    return sum; // (n) is odd
}

int main() {

    // Declaring two vectors, one with an odd length
    // and the other with an even length
    vector<int> arr1 = {1, 2, 3, 4};
    vector<int> arr2 = {1, 2, 3, 4, 5};
    cout << getSum(arr1) << endl;
    cout << getSum(arr2) << endl;
    return 0;
}
```

Output

0

Time Complexity Analysis:

- **Best Case:** The order of growth will be **constant** because in the best case we are assuming that (n) is even.
- **Average Case:** In this case, we will assume that even and odd are equally likely, therefore Order of growth will be **linear**
- **Worst Case:** The order of growth will be **linear** because in this case, we are assuming that (n) is always odd.

Next Articles to Read

- [Asymptotic Notations](#)
- [Analysis of Loops](#)
- [Analysis of Recursive Functions](#)

Reference books:

1. "Introduction to Algorithms" by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein is a comprehensive guide to algorithm analysis, including worst, average, and best case analysis.
2. "Algorithm Design" by Jon Kleinberg and Éva Tardos provides a modern approach to algorithm design, with a focus on worst case analysis.
3. "The Art of Computer Programming" by Donald Knuth is a classic text on algorithms and programming, which includes a detailed discussion of worst case analysis.
4. "Algorithms Unlocked" by Thomas H. Cormen provides a gentle introduction to algorithm analysis, including worst, average, and best case analysis.

Analysis of Algorithms | Set 2 (Worst, Average and Best Cases)

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